Marine Pollution Bulletin 172 (2021) 112858

Contents lists available at ScienceDirect

Marine Pollution Bulletin

journal homepage: www.elsevier.com/locate/marpolbul

Cigarette butts in beach litter: Snapshot of a summer holiday

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ARTICLE INFO

Keywords:
Beach litter
Cigarette butts
Beach users
Sandy beaches
Beach cleaning

ABSTRACT

Cigarette butts are considered the most common form of personal litter around the world. It is estimated that 5.5 trillion cigarettes are consumed globally each year and that 4.95 trillion are discarded in some natural or anthropic environment around the world. This study evaluated the pollution of urban beaches in the Brazilian Northeast by cigarette butts in relation to other types of litter. Samplings were conducted in a month of high season (January 2016), at eight heavily used beaches. In the place with the highest concentration of users in each beach, a 200 m transect (3 m width) was established parallel to the water on the most recent strandline. All items >1 cm were counted and classified into categories according to composition (plastic, cigarette butts, wood, glass, metal, paper and organic, such as food leftovers and coconuts) using a standard worksheet. Cigarette butts were considered as an isolated category of litter. A total of 10,880 items was registered. The most abundant items were plastics (44.96%) and cigarette butts (38.36%). Among plastics, the most abundant items were straws, metallic food packaging, sticks (lollipop and popsicle) and cups. Plastic fragments were also abundant in all beaches evaluated, corresponding to 14.5% of the total plastics.

The intense use of urban beaches is responsible for the generation of large amounts of solid wastes that directly or indirectly affect these environments (Santos et al., 2005a; Simeonova et al., 2017; Becherucci et al., 2017; Silva et al., 2015). The contribution of users is proven and reported in numerous studies (Araújo and Costa, 2006; Becherucci et al., 2017; Dias-Filho et al., 2011; Novotny and Slaughter, 2014; Silva-Cavalcanti et al., 2009; Silva et al., 2008; Silva et al., 2015).

Cigarette butts are considered the most common form of personal litter found on beaches and urban environments around the world. It is estimated that 5.5 trillion cigarettes are consumed globally each year and that 4.95 trillion are inadequately discarded in natural or anthropic environment around the world (Araújo and Costa, 2019a). The cigarette butt is composed of cellulose acetate fibers (Araújo and Costa, 2019b). The polymer is obtained from the cellulose, after the addition of acetic anhydride and acetic acid. Finally, together with the chopped tobacco leaves, it is coated with paper when the cigarette is rolled. The size of the filter usually does not exceed 3 cm and have light color, both internally and externally. Some factors favour its accumulation and permanence in coastal environments (Araújo and Costa, 2019a); the cellulose acetate presents an extremely slow degradation, even under severe weathering conditions (Haynes et al., 1999; Puls et al., 2011).

The presence of cigarette butts on beaches, in addition to the obvious aesthetic impact, also poses risks to the marine biota. This type of residue would be classed as mesoplastics, a transition size between macro and microplastics (Hanvey et al., 2017). This size can pose several threats to the environment and biota; as the ingestion, reported for *Chelonia mydas* and *Eretmochelys imbricata* in a study conducted on the coast of Bahia (Macedo et al., 2011). Although the risk of ingestion is the main impact reported for vertebrate animals, there are also records of impacts for invertebrates. Each discarded filter contains numerous chemical compounds that remain after the burning of the cigarette, such as tar, cyanide, benzene, cadmium and lead, among others; when these substances become widespread in the environment, they become potentially hazardous to aquatic organisms (Araújo and Costa, 2019b; Pon and Becherucci, 2012). Booth et al. (2015), tested the effects of a solution prepared from cigarette filters, at different dilutions and in an undiluted treatment, on the behavior and survival of three mollusk species (Gastropoda). The results indicated changes in the behavior of the individuals evaluated (as in the ability to move in response to desiccation stress), even at the lowest concentrations. Regarding survival, 100% mortality was observed in the treatment without dilution; however, even in diluted solutions there was mortality, showing the high degree of toxicity of the compounds present in the filter.

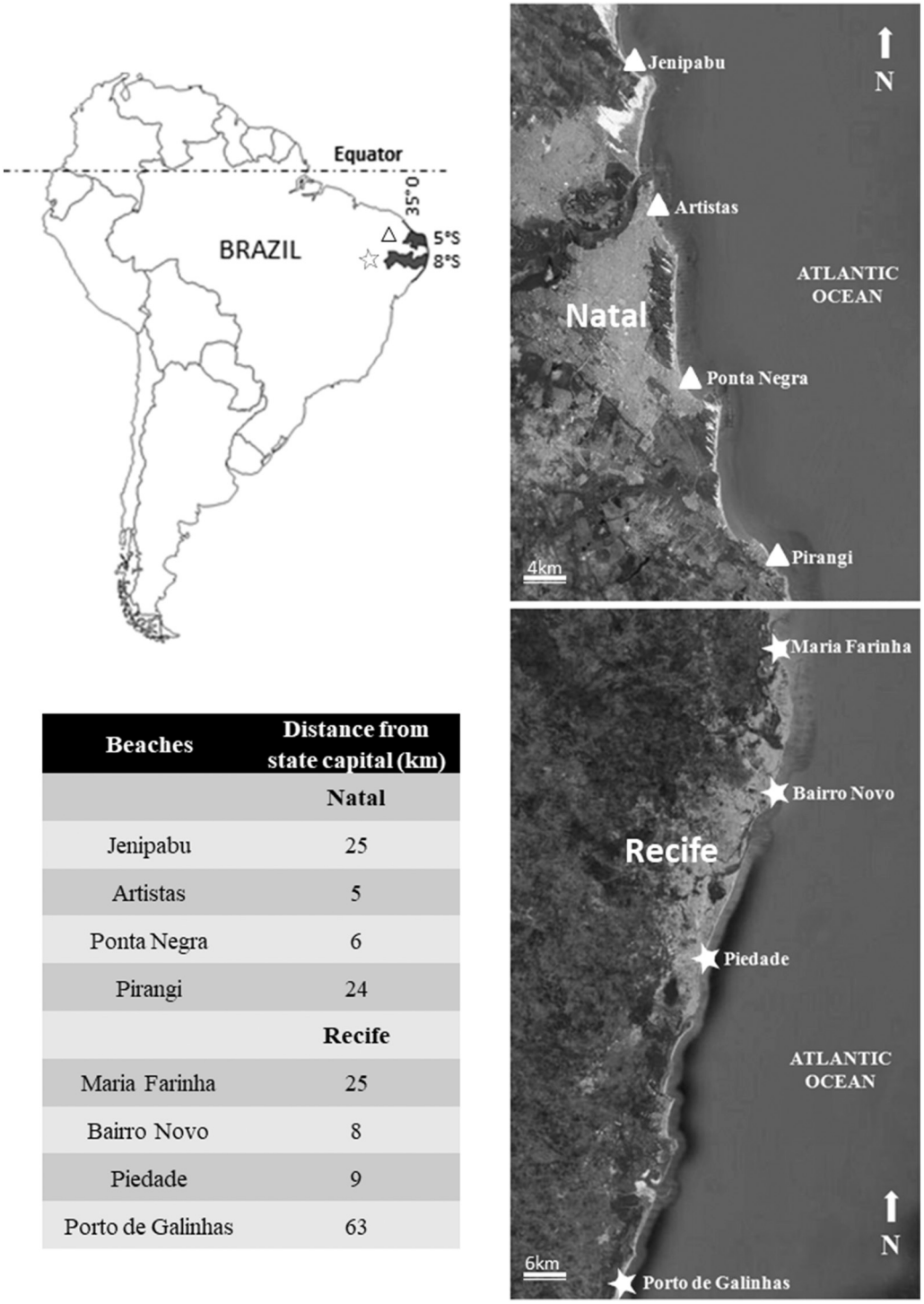
In addition of being easily ingested by many vertebrates and

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https://doi.org/10.1016/j.marpolbul.2021.112858
Received 23 March 2021; Received in revised form 3 August 2021; Accepted 11 August 2021
Available online 19 August 2021
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invertebrates, it is a potential source of secondary microplastics in the form of fibers. Fragmentation can occur both in sand and in seawater, depending on the prevailing environmental conditions (sunlight, desiccation, and abrasion). Also, filters could interact with the biota through bioturbation, and become more disperse and/or buried in the sand. Beach scavengers as *Ocypode quadrata* (ghost crab) can carry cigarette butts and filters down their holes, a likely process that exemplify unexpected interactions that could become relevant when considering the wide availability of this pollutant (Costa et al., 2018, 2019).

This study evaluated the pollution by cigarette butts in relation to other types of litter, at eight beaches in the Northeast of Brazil; four in the state of Pernambuco (Maria Farinha, Bairro Novo, Piedade and Porto de Galinhas) and four in the state of Rio Grande do Norte (Jenipabu, Artistas, Ponta Negra and Pirangi) (Fig. 1). Among them, there are two beaches that stand as important touristic assets for these States. Porto de Galinhas has natural pools protected by reefs, which at low tide are exploited in shallow dives and raft trips, as well as presenting infrastructure for tourism. Jenipabu has high dunes, which are explored by buggy and dromedaries' rides, calm sea, and warm waters. This is the first study about cigarette butts as a singular beach litter fraction at



Beaches	Distance from state capital (km)
	Natal
Jenipabu	25
Artistas	5
Ponta Negra	6
Pirangi	24
	Recife
Maria Farinha	25
Bairro Novo	8
Piedade	9
Porto de Galinhas	63

Fig. 1. Location of study areas in Rio Grande do Norte (Δ) and Pernambuco (☆) states.
Source: Google Earth.

beaches in this region and, for some of then, the first beach litter study ever.

The distances from the state capitals to the studied beaches varied from 5 to 63 km (Fig. 1). The eight studied beaches were classified into two of the three development levels proposed by Araújo and Costa (2008). Among the beaches included in the present study, two had already been analysed by those authors. In Araújo and Costa (2008), Maria Farinha was considered a "developed" beach, ranking "good" for the environmental indicator and "regular" for the socioeconomic indicator. Bairro Novo was considered "overdeveloped" and ranked "regular" for both indicators.

These beaches were chosen because they are heavily used by residents, day-visitors and tourists in the high season (November to February), despite differences in the degree of urban development of each municipality. Usually, the peak season is the most critical for these coastal locations, when the presence of a greater number of users increases the demand for water, services and infrastructure; in addition to increasing sewage production of mainly solid wastes (Sunlu, 2003; Padilla, 2015).

Samplings were conducted in high season (January 2016). In the place with the highest concentration of users in each beach, a 200 m transect (3 m width) was established parallel to the water on the most recent strandline (Davies and Gillham, 2004). All items >1 cm were counted and classified into categories according to composition (plastic, cigarette butts, wood, glass, metal, paper and organic, such as food leftovers and coconut shells). Cigarette butts were considered as an isolated category of litter (Becherucci et al., 2017; Smith et al., 2014), and also because of its specific composition (cellulose acetate).

A total of 10,880 items was registered (in 4800m^2), of which 5519 were on beaches in Pernambuco and 5361 on beaches in Rio Grande do Norte. The most abundant items were always plastics (4892 items), and cigarette butts (4174 items).

Cigarette butts were always higher than 30% of beach litter items, reaching a maximum value of 44.9% (Fig. 2). This percentage did not vary widely, despite beaches being different in terms of distance from the main urban centers, socioeconomics of users and municipality development levels.

Among plastics, four types of residues were most abundant: drinking straws, metalized food packaging (BOPP-Polypropylene Bioriented), sticks (lollipop and popsicle) and cups, accounting for 63.6% and 55.5% of total plastics, in Pernambuco and Rio Grande do Norte, respectively. For all these items, the most probable source are beach users who litter the environment during their presence (Silva et al., 2008).

Plastic fragments were also abundant in all beaches evaluated, corresponding to 14.5% of the total plastics. This number is high, when it is considered that only items larger than 1 cm were registered (limit of visual accuracy); there is probably a much higher amount of smaller fragments mixed in the sand, due to the high rate of fragmentation to which plastics are subjected when exposed to environmental conditions (Costa et al., 2010). Fauziah et al. (2015) investigated the abundance of small plastic debris buried in the sands at six sampling sites on Malaysian beaches and found a total of 2542 pieces of small plastic debris (the most predominant size was more than 4.75 mm); results indicated the relative abundance of small plastic debris buried in 5 cm deep sand of the selected beaches.

Regarding the other evaluated categories, wood and organic items (food scraps and coconuts) were the most representative, corresponding to 6.69% and 5.46%, respectively. Among the wooden items, skewers and popsicle sticks were the most abundant. Food items usually undergo rapid degradation; however, they serve as a food resource for detritivores and a substrate for the proliferation of microorganisms in the beach sand (Zuza-Alves et al., 2016), causing risks for users.

The density of total litter items was co-related to the density of cigarette butts. Beaches with similar use and maintenance patterns were relatively close to each other on this plot (Fig. 3).

Considering similar distances in relation to the nearest urban center, the beaches of Ponta Negra and Piedade show different patterns in the amount of waste generated when compared to Bairro Novo and Artistas. Ponta Negra and Piedade had the smallest amounts of waste, including cigarette butts. Although relatively close to, or within the state capital (Fig. 1), these beaches show different use patterns in relation to users' socioeconomic profile and income. It is expected that individuals with higher incomes will have better access to formal education, greater concern with environmental quality and conservation of public spaces (Santos et al., 2005b).

However, our results suggest that an important factor interfering with the amount of litter on beaches is the municipal cleaning service (Table 1). Piedade and Ponta Negra have around the clock effective and permanent cleaning plans, through street sweepers that work along the beach and complementary motorized systems that operate every evening. Furthermore, both beaches are subject to coastal erosion, with reduction of the beach environment and possibly of beach litter accumulation.

Bairro Novo and Artistas have only manual daily municipal cleaning services but had higher amounts of waste them Maria Farinha and Pirangi, which are more distant from the capitals of the respective states and without municipal cleaning services. It is possible that Bairro Novo and Artistas being closer to the capital (Fig. 1) and concentrating lower income users allows for a combination of factors that result in greater waste generation.

Porto de Galinhas and Jenipabu, although more distant from the state capitals (Fig. 1), stood out in the amount of litter found; probably due to the numerous attractions that make them the primary postcards of their states. Despite receiving a high number of visitors, neither of these beaches has a consolidated cleaning system or plan.

The presence of users contributes to a proportional increase in beach

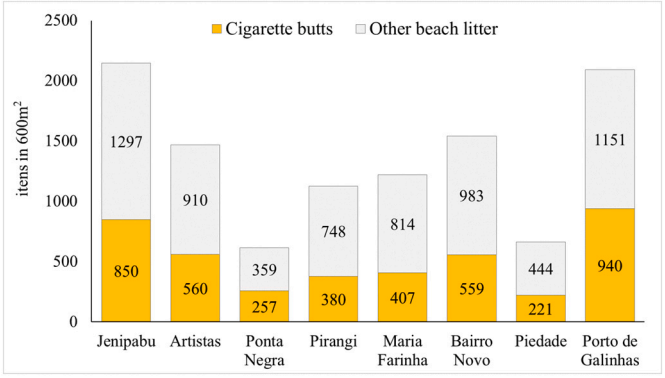


Fig. 2. Cigarette butts in relation to other beach litter items in 600 m^2 transects deployed along the strandline of eight beaches in January 2016.

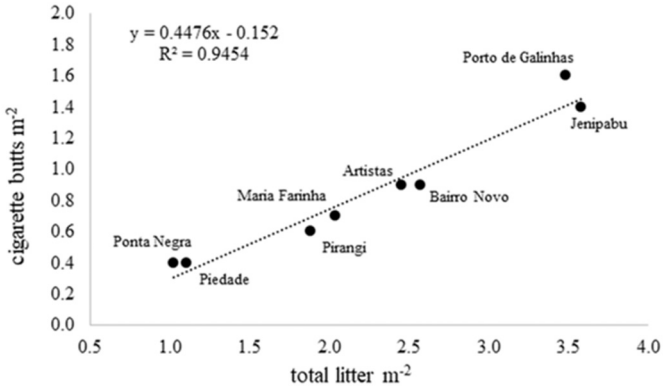


Fig. 3. Correlation between the total items and cigarette butt densities (no. m^{-2}) at the eight beaches studied.

Table 1
Selected studies comparing the amounts of litter and cigarette butts at different beaches, subject to various patterns of use, and the results of the present work. *According to Araújo and Costa, 2008; **According to each author.

Beaches	Beach use characteristics	Total items	Cigarette butts	Author
Ponta Negra; Piedade,	Overdeveloped beach* Manual and mechanic daily municipal cleaning services	1281	478	Present study
Maria Farinha; Pirangi	Developed beach* No municipal cleaning services	2349	787	
Bairro Novo; Artistas	Overdeveloped beach* Manual daily municipal cleaning services	3012	1119	
Jenipabu; Porto de Galinhas	Developed beach * No municipal cleaning services	4238	1790	
Itaipu (Rio de Janeiro – Brazil)	A more intense trade; Several kiosks on the strip of sand; No bins on the beach or a regular and efficient collection of garbage**	18,127	3226	Silva et al. (2015)
Itacoatiara (Rio de Janeiro – Brazil)	Kiosks are located only on the boardwalks**	9245	1671	
Boa Viagem (Pernambuco – Brazil).	Vertical seafront; Permanent residences High presence of users whole year**	165,882	24,803	Silva-Cavalcanti et al. (2013)
Center; Mar Chiquita (Mar del Plata – Argentine)	Very popular beaches with high presence of users**	7394	2539	Becherucci et al. (2017)
Paseo; Faro Querandí (Villa Gesell – Argentine)	Quiet beaches with low presence of users**	2226	762	
Debeli Rtic; Simonov Zaliv; Portoroz (Slovenia)	Beaches classified as touristic (areas of high urbanisation and visit)**	4810	1521	Laglbauer et al. (2014)
Jadranska; Bele Skale; Seca (Slovenia)	Beaches classified as non-touristic (limited or absence of urbanisation and visit)**	1060	167	

commerce. The diversity of items on sale also increases (Araújo et al., 2012). The higher the number of users and the length of stay, the greater the generation of residues, represented mostly by plastic packaging and cigarette butts, which are known to have slow degradation (Araújo et al., 2012; Puls et al., 2011; Silva-Cavalcanti et al., 2009; Silva et al., 2008). The correlation between the concentration of users and the occurrence of marine litter has also been reported for other Brazilian beaches, especially in the northeastern and southern regions (Araújo and Costa, 2007; Silva-Cavalcanti et al., 2013; Santos et al., 2005b).

During the high season there is an increase in littering (Becherucci et al., 2017; Leite et al., 2014; Santos et al., 2005b), suggesting this period could be considered an extreme condition for plastics and cigarette butts' pollution. It could set the limits of the environmental support capacity, a useful tool for managers and formulation of public policies.

According to Kuo and Huang (2014), the presence of litter on sandy beaches is significantly related to number of visitors. People dispose wastes irregularly, especially because of the absence or limited quantity of containers for the collection near the places of use on the beach. In the beaches evaluated in the study, there were no adequate or sufficient infrastructure for waste collection, such as bins in the sand area, where most users concentrate. In the case of cigarettes, the habit of light it off in the sand and leaving it immediately there and/or bury it, still prevails; perhaps as the result of an old habit or as an attempt to hide the misdeed.

The large amount of cigarette butts found during the present work reflects global trends (Araújo and Costa, 2019a). According to the Ocean Conservancy, most of the world's beaches are littered with cigarette butts. In the 2015, 2016, 2017 and 2018 reports (ICC reports: https://oceanconservancy.org/trash-free-seas/international-coastal-clean up/annual-data-release/), cigarette butts ranked first in number of waste items collected, with amounts greater than 1,800,000 items annually found. In Brazil, only in 2015, the campaign collected 39,606 items of litter, 6311 of which were cigarette butts.

Ariza et al. (2008) assessed two urban beaches (located in the main nucleus of the municipality) and two urbanized beaches (located in residential areas outside the main nucleus of the municipality). The total production and deposition density of waste on the urban beaches were clearly higher than on urbanized beaches. Small items of litter as cigarette butts tended to accumulate at the peak of the season. According to them, urban beaches have more users than other types of beaches, such as urbanized or natural beaches, and consequently, yield larger quantities of wastes.

Eight beaches on Bulgarian Black Sea coast were selected to evaluate marine litter by Simeonova et al. (2017), according to the following criteria: city beaches with intense recreational activities; beaches near uninhabited or less populated and beaches near rivers. The highest densities of marine litter occurred on the beaches used for recreation activities especially during summer period. A total of 19,798 were collected and cigarette butts were the first most frequently found type of residues, with a total of 4514.

Silva et al. (2015) also observed that the number of users influenced the amount of waste, increasing the number of items found. Several other studies also report the high incidence of cigarette butts on beach sand and the largest quantities were found on the most used beaches (Table 1).

Numerous factors converge to determine the highly abundant presence of this type of waste in the beaches. Smoking is widespread throughout society, and the irregular disposal of cigarette butts in public spaces in cities is reflected in coastal environments. Cigarette butts can be transported by water and wind to the beaches and join those that are directly discarded by users in this environment (Araújo and Costa, 2019a).

Beach cleaning is usually manually, using a rake and large sieves for transport; sometimes small mechanized vehicles are also used in limited dry sand areas (Araújo and Costa, 2019a). All debris is then bagged, and road transported. According to Widmer and Reis (2010), beach cleaning is relatively inefficient for small items, especially cigarette butts, seals, bottle caps, and fragments. Nevertheless, when cleaning is done manually, the focus is usually the most recent strandline, where small items (*meso* litter) are hauled together with wrack and other items, which can contribute to reducing the total number of these items accumulated on the beaches.

All beaches studied presented unacceptable amounts of litter, especially because they are places with high frequency of visitation. Beach pollution might impair use and consequently compromise its use as a source of income for local sellers and municipalities. As litter is the most visible type of pollution, the beach's aesthetic becomes compromised which can undermine tourism and negatively impact the local economy. In addition, there is always the risk for marine biota.

The high number of cigarette butts found suggests its use as an indicator item of the contribution of users in the pollution of urban

beaches. The use of flag items reduces the time spent with surveys of all items, facilitating monitoring of more areas. Research that requires fieldwork is costly. It takes time, displacement costs and other needs, especially when involving many people. The larger and more diverse the sample area, the higher the costs. The collection effort should be planned with the objective of optimizing both economic and human resources. We also recommend that beach litter surveys are accompanied of other variables that can help understanding beach use patterns and management.

Due to the proven difficulty of eliminating the generation of this type of waste, it is necessary to implement measures to reduce cigarettes consumption. The decrease in consumption can bring innumerable benefits directly related to human health. Campaigns aimed at smokers, focusing on disposal to reduce accumulation in the environment are also necessary. Proper disposal can benefit the environment by reducing pollution risks to the biota.

CRediT authorship contribution statement

MCBA was responsible for the conception of the work, field campaigns and manuscript development; MFC participated in manuscript development.

Funding

Funding was provided by CAPES and CNPq, Brazilian Government. MFC is a CNPq Fellow.

Declaration of competing interest

The authors declare that there are no competing interests regarding this work.

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